Università di Genova

DIBRIS: Robotics Engineering

AIRO2 PRESENTATION

ASSIGNMENT D5-V2: AGRICULTURAL ROBOTICS - IRRIGATION PLANNING

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PROBLEM DEFINITION AND OBJECTIVES

Problem: one robot is deployed in a field divided in crops, as shown in figure 1.
It must be able to bring all the crops to a certain moisture level, assumed as stable, while keeping count of environmental constraints and limited resources

Assumptions:

- robot always starts in start1;
- locations connected in the four directions;
- moisture level represented in numbers, 50 is the stability threshold

Objective: define solution plans using features from basic PDDL and PDDL+ to:

- use the planner's capabilities;
- explain the course of action that brings to an approximation of realistic scenarios.

Planner used: ENSHP (online and local)

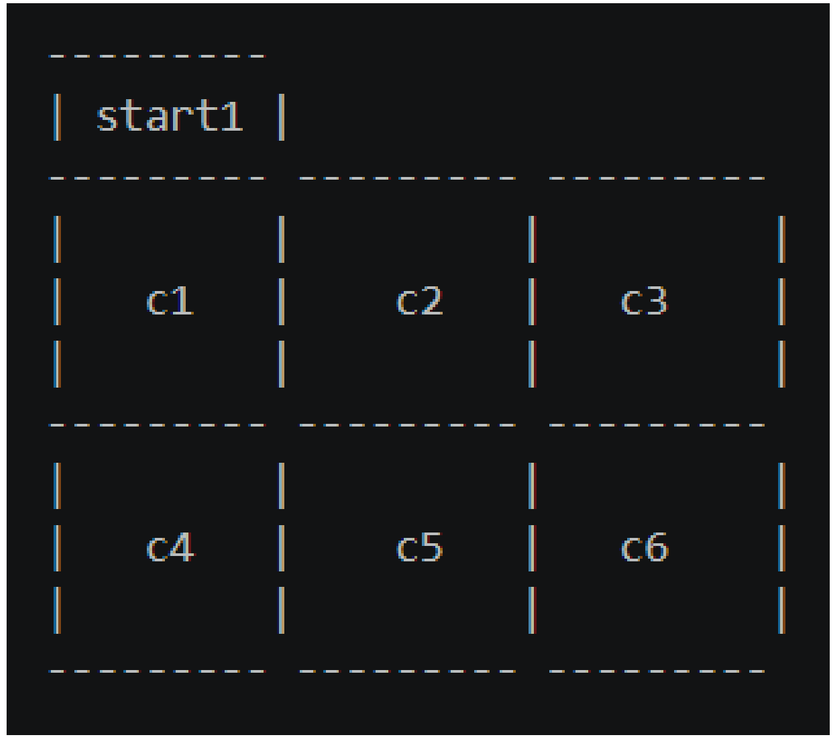


Fig 1: Testing environment

BASIC PDDL DOMAIN

Domain file common to both the abundant and limited resource problem files.
No environmental dynamics involved, used to define the robot's reasoning logic.

```
(:action check_levels
    :parameters (?c - crop)
    :precondition (and
        (forall (?other - crop) (>= (moisture_level ?other)
        (moisture_level ?c)))
    )
    :effect (and
        (is-priority ?c)
        (forall (?other - crop)
            (when (not (= ?other ?c)) (not (is-priority ?other)))
        )
    )
)
```

Fig. 2: code structure of priority assigment

Main actions:
1) *Movement*: used to change the robot's position in the field. Allowed between two connected crops.
2) *Check_levels*: checks every crop's moisture level and chooses the one with the lowest level.
3) *Irrigate*: when the robot is located on a prioritized crop, it irrigates that crop by decreasing its water supply by 10 and increasing that crop's moisture level by 10. Numeric fluents avoid binary representation.
4) *Sacrifice*: a crop is flagged as sacrificed when the robot cannot irrigate it due to complete resource delpetion.

Robot logic: Choose priority crop → Reach it → Irrigate it → Repeat

PDDL+ DOMAIN

PDDL+ allows, by introducing continuous time, to add more realism in the current scenario.

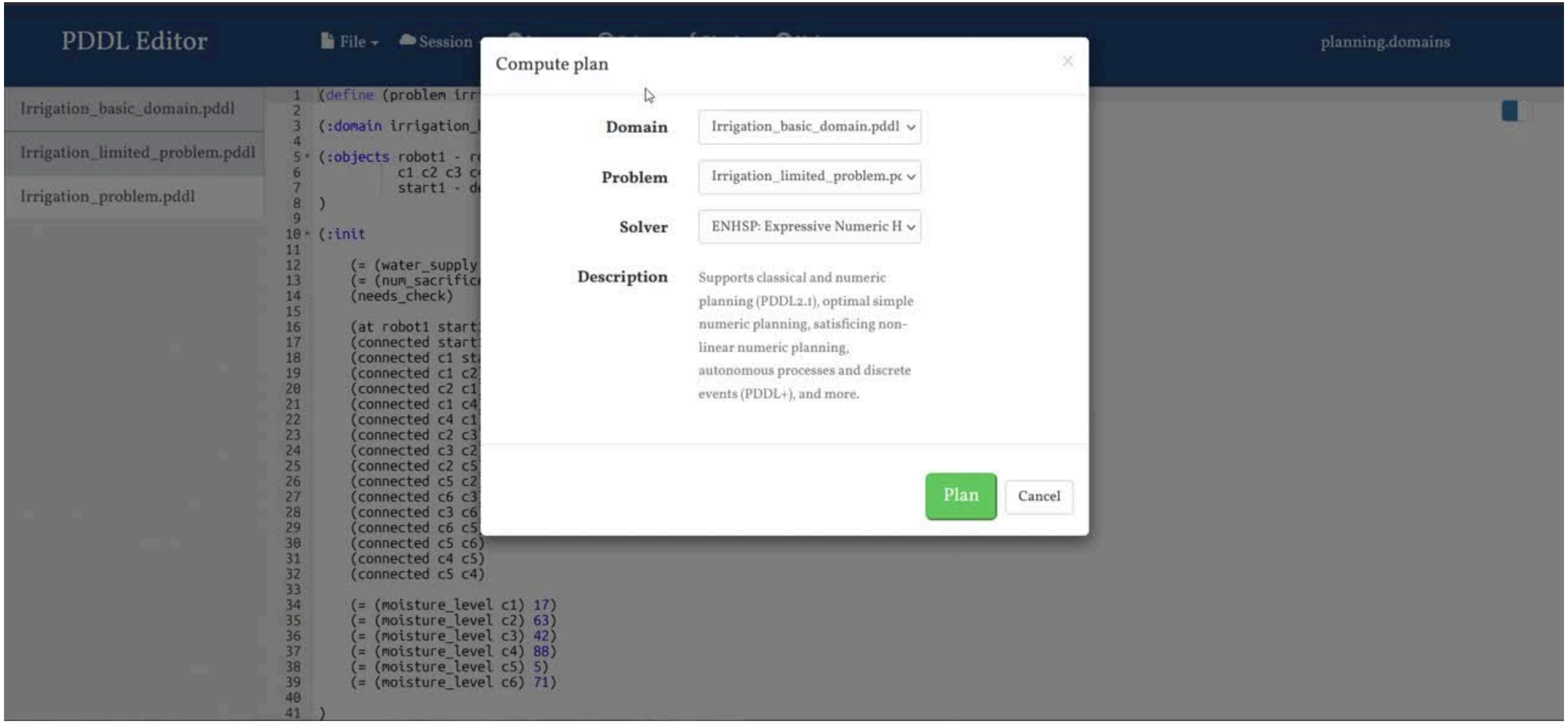
Two main modifications emerge when compared to the basic scenario:

- introduced a process which passively drains every crop's moisture level and an event which makes a crop unusable if left alone for too much time;
- modified the logic that regulates movement and irrigation: they are no more instant actions, but processes toggled by actions and events;
- modified the priority logic to make it less computationally heavy in continuous time.

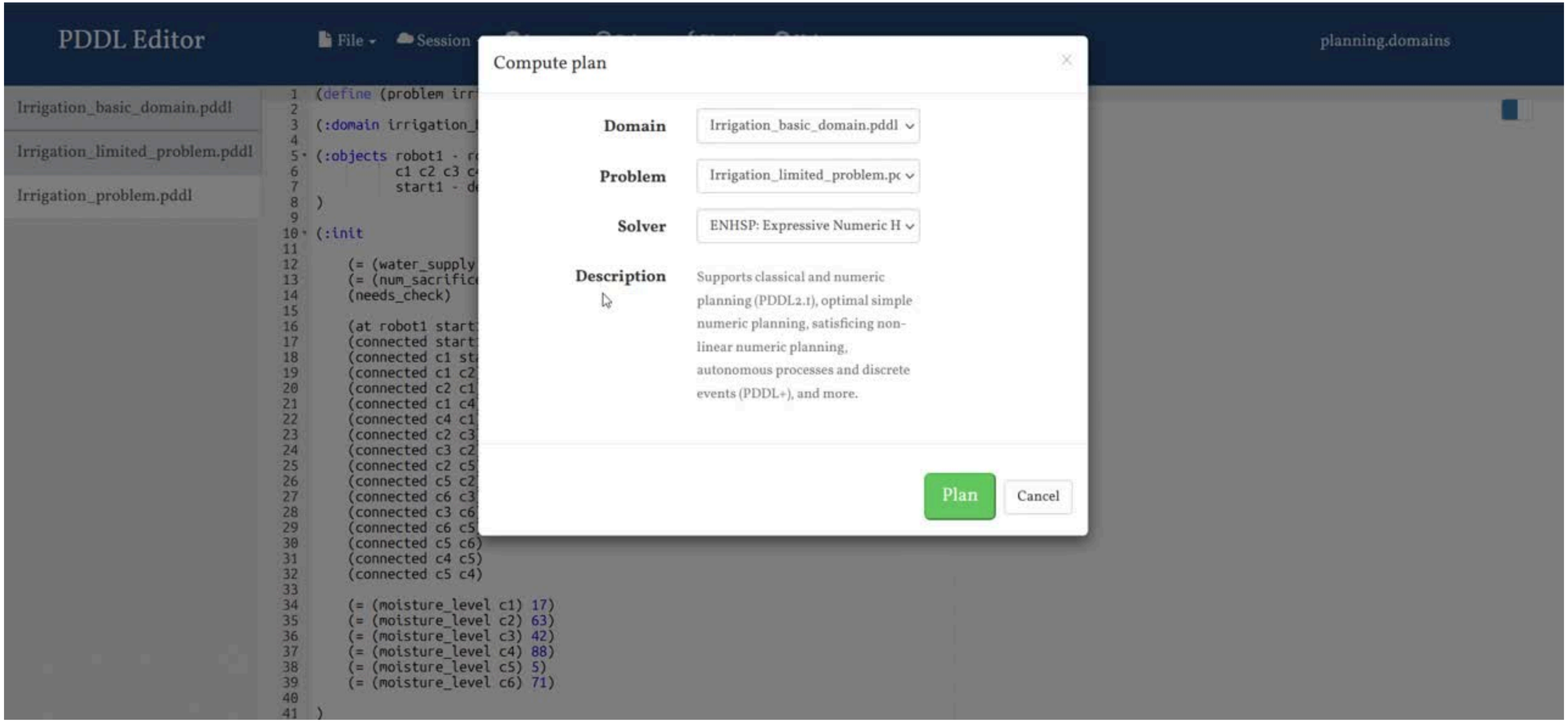
```
(:process evaporation
    :parameters (?c - crop)
    :precondition (and (>= (moisture_level ?c) 10) (not (unusable ?c)))
    :effect (and
        (decrease (moisture_level ?c) (* 1 #t))
    )
)

(:event drought
    :parameters (?c - crop)
    :precondition (and
        (<= (moisture_level ?c) 10)
        (not (unusable ?c))
    )
    :effect (and
        (assign (moisture_level ?c) 0)
        (unusable ?c)
        (increase (num_drought_events) 1)
    )
)
```

Fig.5: environmental conditions implemented

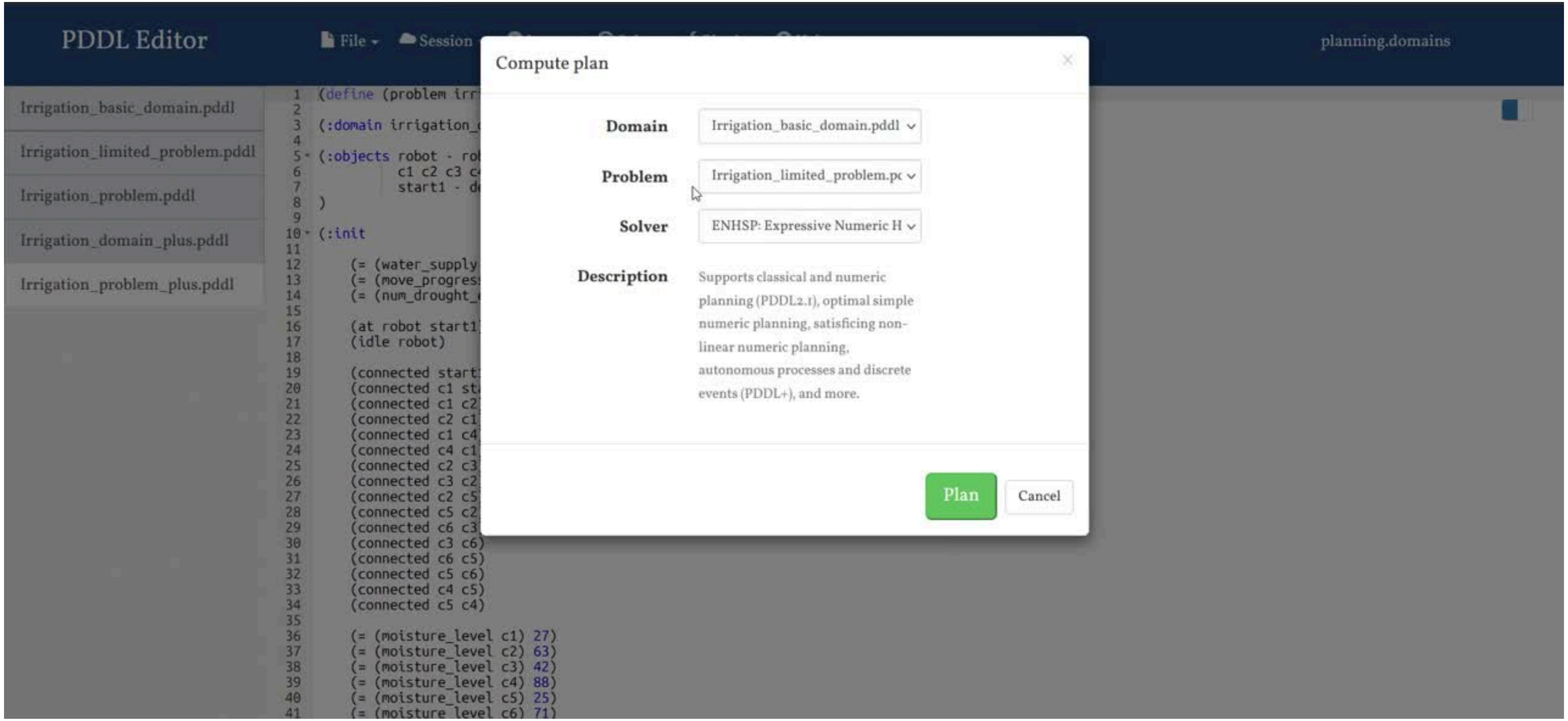
RESULTS - ABUNDANT WATER PROBLEM



RESULTS - LIMITED WATER PROBLEM



RESULTS - PDDL+



CONCLUSIONS

Main challenges:

- Implementing the priority logic and solving related bottlenecks in both scenarios
- Conversion of movement and irrigation actions in continuous time
- Ordering the actions in both scenarios to make the course of action logically sound

Possible differences from current implementation:

- Priority logic in basic problems is fired after every irrigation step, rather than after a complete irrigation. Possible to modify it since no environmental dynamics are involved.
- In PDDL+, movement is regulated by a action → process → event logic, while irrigation has a action → process → action logic.

Limitations:

- All problems' solvability is hard-constrained by the water supply's quantity
- Known issue: first action in PDDL+ fires after two seconds